

DiffAudit: Evidence-Contracted Membership Leakage Auditing for Diffusion Models

Anonymous Authors

Abstract—Membership inference against diffusion models is often reported as a scalar score, yet a deployable privacy audit needs more than a high AUC. We treat the measurement unit as a claim-bound evidence packet: a fixed target model, member/nonmember semantics, row-level score or response coverage, metric provenance, consumer boundary, and surface-delta evidence for portability. We present DiffAudit, an evidence-contracted audit methodology for diffusion-model membership leakage. DiffAudit separates admitted evidence from research candidates and watch items, reports finite-tail low-FPR metrics with their packet denominators, and leaves attractive results non-admitted when their artifacts do not support the claimed audit semantics. On the current evidence bundle, DiffAudit admits five rows under black-box, gray-box, and white-box contracts, spanning reconstruction risk, PIA, a stochastic-dropout comparator, GSA, and a DPDM defense comparator. We also report a controlled H2 response-cloud case study: an output-output geometry scorer reaches AUC 0.9615 on an existing response cache and survives label-shuffle, seed-offset control, seed-stability, and same-family cross-cache transfer checks, but fails to cleanly port to an SD/CelebA img2img contract. Finally, bounded scouts and a frozen metadata-only search batch over GitHub and arXiv show why second-asset claims require artifact contracts rather than metric prose: 17 fixed-search rows add selection-process evidence but no new admitted audit row. DiffAudit is therefore a measurement framework: it exposes verified leakage where evidence supports it and marks portability claims as non-admitted where the public artifact surface is incomplete.

I. INTRODUCTION

Diffusion models can memorize or overfit training examples in ways that enable membership inference attacks. The literature has produced strong black-box, gray-box, and white-box signals, including reconstruction similarity, posterior-error statistics, proximal initialization, conditional likelihood discrepancy, reconstruction-based black-box attacks, and gradient-based attacks [1], [2], [3], [4], [5]. However, paper-level scores are not automatically audit evidence. A model owner, regulator, or downstream runtime needs to know which target checkpoint was audited, what row IDs define members and nonmembers, whether scores exist per sample, how low-FPR tails should be interpreted, and which claims are blocked by missing artifacts.

DiffAudit is built around this distinction. We treat the measured object as an evidence packet, not as a method name or a single score. A packet is admissible only when it is connected to a contract that records target identity, split semantics, score or response coverage, metric provenance, consumer boundary, and the surface delta needed for portability claims. This paper argues that such evidence contracts are not paperwork; they are scientific controls. They expose source-label confounding,

prompt-only shortcuts, finite-tail overclaiming, unsupported consumer promotion, and the common failure mode where a high-level metric cannot be replayed or consumed by a real audit system. The measurement instrument is therefore the gate outcome itself: it records which claim survives, which claim is blocked, and which missing surface caused the boundary.

The audit decision problem has three consumers. A model owner needs to know whether a deployed or released checkpoint creates measurable membership risk under a stated access mode. An independent auditor needs row-bound artifacts that allow metric replay and low-FPR interpretation without retraining the target. A downstream runtime or report generator needs machine-checkable boundary language so that candidate mechanisms do not become user-facing audit claims. DiffAudit is therefore not an attack-SOTA paper: it studies when a diffusion MIA result is safe to consume as audit evidence.

Our current evidence supports three findings. First, a small admitted bundle can already demonstrate meaningful diffusion privacy risk across access levels: black-box reconstruction risk reaches AUC 0.837, gray-box PIA reaches AUC 0.841, and white-box GSA reaches AUC 0.998 under their stated contracts. Second, a new H2 response-cloud analysis suggests that repeated output geometry contains membership signal beyond simple seed-to-output distance. This signal is strong and controlled inside the H2 response-strength setting, but it remains a research candidate because the SD/CelebA img2img portability check is weak or not distinct from simpler distance features. Third, many plausible second-asset routes are scientifically useful precisely because they fail: ReDiffuse STL-10 has a clean split and scoreable short target but random-level metrics; CommonCanvas has a real response packet but weak scorers; Tracing the Roots has positive feature tensors but no raw image-level consumer contract.

The paper therefore makes the following contributions:

- We define the claim-bound evidence packet as the measurement unit for diffusion membership auditing and specify six gates that separate admitted, candidate, support-only, watch, and negative evidence.
- We report a five-row admitted evidence bundle with unified AUC, ASR, TPR@1%FPR, TPR@0.1%FPR, cost, provenance, and boundary language.
- We introduce and evaluate an output-cloud geometry case study that survives several bounded controls while remaining honestly limited to its current response-contract family.
- We document negative and support evidence from public artifacts, showing how missing checkpoints, broken

row bindings, weak transferred signals, and consumer-boundary mismatches block overbroad portability claims.

II. RELATED WORK

A. Membership Inference and Evidence

Membership inference was introduced for discriminative machine-learning services as a way to infer whether a record was used during training [6]. Subsequent work connected the risk to overfitting and clarified the statistical interpretation of membership signals [7], [8]. Recent position work further argues that membership inference should not be treated as standalone proof that a model was trained on a particular datum [9]. DiffAudit adopts this caution as an audit-design principle: a high score is useful only after its target, split, metric, and consumer boundary are explicit.

B. Diffusion Memorization and MIA

Diffusion models expose privacy risk both through memorized generations and through membership signals. Extraction and replication studies show that training examples or near-duplicates can appear in model outputs under some conditions [10], [11]. Diffusion MIA work then studies more direct member/nonmember decisions under different access levels. SecMI measures posterior-estimation error across denoising steps [1]; PIA reduces gray-box query cost through proximal initialization [2]; CLiD uses conditional likelihood discrepancy for text-to-image models [3]; ReDiffuse studies black-box variation-query attacks [12]; and GSA uses gradient statistics as a strong white-box signal [5]. DiffAudit does not replace these methods. It asks which method outputs can be safely compared, replayed, and consumed as audit evidence.

C. Reporting and Reproducibility Artifacts

DiffAudit is also related to broader reporting and reproducibility work in machine learning. Reproducibility checklists and the NeurIPS reproducibility program emphasize reporting the data, code, hyperparameters, and evaluation conditions needed to interpret machine-learning results [13]. Datasheets and model cards similarly argue that downstream consumers need structured documentation about the origin, intended use, and limitations of datasets and models [14], [15]. DiffAudit adapts this documentation logic to diffusion privacy audits: the artifact is not only a model or dataset, but a row-bound membership-evidence packet with an explicit consumer boundary.

III. AUDIT SURFACES

Diffusion MIA results are not interchangeable because they observe different audit surfaces. A black-box response packet can support an output-consumer claim if target identity and query rows are fixed. A gray-box trajectory or posterior-statistics packet can support a mechanism or evaluator claim, but requires a different consumer boundary. A white-box gradient packet is useful as an upper-bound risk comparator, not as a black-box report. Feature packets and collaborator-local scores can be scientifically useful while remaining support-only. Table I records the surfaces used in this paper.

TABLE I
AUDIT SURFACES AND ALLOWED EVIDENCE ROLES.

Surface	Allowed role in DiffAudit
Black-box responses	Admitted only with fixed target, split, response coverage, and metric replay.
Gray-box trajectory/statistics	Admitted or candidate depending on provenance and consumer semantics.
White-box gradients	Upper-bound comparator unless the audit consumer has white-box access.
Feature packets	Support-only unless raw target/sample provenance and regeneration assets are available.
Collaborator/local packets	Stress tests unless public row binding and source-confounding checks pass.

IV. EVIDENCE CONTRACT

DiffAudit uses six gates before a result can support a portability or runtime audit claim. The target identity gate requires a fixed model architecture, checkpoint identity, training provenance, and accessibility. The split contract requires precise member and nonmember manifests. The score or response gate requires row-level coverage and labels. The metric gate requires AUC, ASR, TPR@1%FPR, TPR@0.1%FPR, nonmember denominator, and traceability to a score array or response packet. The consumer-boundary gate requires a stated consumer and allowed claim language. The surface-delta gate checks whether a claimed second asset actually differs in model, data, modality, or response contract rather than repeating the same family.

The gates are intentionally stricter than ordinary artifact availability. A repository can contain code without a fixed target checkpoint, a checkpoint can exist without immutable member/nonmember manifests, a score file can be scoreable without proving the same consumer boundary, and a strong candidate can still be unsafe to expose as a runtime audit row. DiffAudit therefore asks what claim an artifact supports, not whether an artifact is globally useful or whether the original paper is reproducible in every detail.

Low-FPR metrics are reported as finite empirical packet readouts. For example, TPR@0.1%FPR on 100 nonmembers has a false-positive budget of 0.1 and should not be described as a calibrated continuous sub-percent operating point. This rule is important because many diffusion MIA packets are small, and strict-tail claims can otherwise sound stronger than the evidence permits.

The consumer-boundary gate is the most conservative part of the contract. It records who may consume the result and what language must accompany it. A white-box gradient attack may be a valid upper-bound comparator but not a black-box product result; a feature-packet replay may be useful evidence for temporal trajectory leakage but not a raw image audit; a cross-source Stable Diffusion packet may be a stress test but not same-distribution membership evidence. These distinctions are enforced before writing the paper, not added as defensive caveats after a claim is made.

TABLE II
DIFFAUDIT EVIDENCE-CONTRACT GATES.

Gate	Question answered
Target identity	Which model/checkpoint is being audited?
Split semantics	Which rows are members and nonmembers?
Score/response coverage	Is there row-level evidence for every sample?
Metric provenance	Can headline metrics be traced to rows or frozen responses?
Consumer boundary	Who may consume the result and as what claim?
Surface delta	Is a claimed second asset actually non-adjacent?

TABLE III
EVIDENCE STATES USED BY DIFFAUDIT.

State	Meaning
Admitted	All gates pass for the stated consumer boundary; row may appear in audit reports.
Candidate	Scientifically useful signal, but at least one portability or consumer gate remains open.
Support-only	Useful context or mechanism evidence, but not a direct benchmark row.
Watch	Public signal worth tracking, with no row-bound evidence yet.
Negative	Bounded route-selection result that closes or narrows a hypothesis.

V. MEASUREMENT PROTOCOL

DiffAudit stores each result as one of five states. An admitted row can be reported to downstream consumers under its stated boundary. A candidate row is scientifically useful but not consumer-ready. A support-only row can motivate or contextualize a mechanism but cannot serve as a direct benchmark. A watch item is tracked for future artifacts. A negative row answers a bounded decision question and usually closes a low-value expansion.

Metrics are recomputed from row-level scores or frozen response packets when available. We report AUC, attack success rate (ASR), $\text{TPR}@1\%\text{FPR}$, and $\text{TPR}@0.1\%\text{FPR}$, but we do not treat all four values equally. Low-FPR values are interpreted through the nonmember denominator: on a packet with 25 nonmembers, $\text{TPR}@0.1\%\text{FPR}$ has no continuous sub-percent meaning. This is why DiffAudit records both metrics and packet boundaries instead of publishing a leaderboard.

A. Evidence States

The evidence state is part of the measurement result. It is not a project management label. The same numerical score can be admitted in one setting, candidate-only in another, and support-only in a third if its target identity, split, row coverage, or consumer boundary changes.

B. Metric Replay

DiffAudit treats metric replay as a provenance check. When a score array exists, metrics are recomputed from row labels rather than copied from prose. When only responses exist, the

scorer and response manifest must be fixed before the metric can be used. For a nonmember packet of size n_0 , FPR takes values on a discrete grid of $1/n_0$. Thus $\text{TPR}@0.1\%\text{FPR}$ is a strict empirical tail whenever $0.001n_0 < 1$. We still report the readout because it is useful for comparing packets, but the text must retain the denominator and avoid calibrated-risk language.

Table IV summarizes how current non-admitted routes are used in the paper. The matrix is deliberately coarse: it is a claim-control device, not a new scoring system.

VI. CLAIM-CONTROL CORPUS

The current claim-control corpus is intentionally small but heterogeneous. The admitted bundle contains five rows that already pass the consumer boundary. H2 output-cloud geometry contributes a mechanism candidate with strong internal controls. ReDiffuse STL-10, CommonCanvas, and MIDST contribute bounded negative results because each answers a specific route-selection question. Tracing the Roots contributes positive feature-packet evidence, while the collaborator Stable Diffusion ReDiffuse packet contributes a source-confounding stress test.

This corpus is not a broad survey of every diffusion MIA artifact. It is the evidence surface available under strict row-binding and no-overclaim rules. The counts below are diagnostic examples for claim control, not estimates of public artifact prevalence. The next expansion should freeze inclusion criteria, check public target identity, inspect split and score/response availability, and only then decide whether any execution is justified. This keeps the measurement paper from turning artifact triage into uncontrolled engineering work.

A. Corpus Protocol

The current corpus was built from DiffAudit’s audited evidence surface. A standalone artifact-claim paper would require a larger frozen corpus or a second independent label review before making aggregate claims. The protocol is deliberately lightweight: freeze the search date, sources, and keywords; record public metadata only; label each artifact with the six gates; and separate public artifacts from internal controls and collaborator-local packets. No large model or dataset download is justified until the metadata gate shows that a row-bound claim could change.

We instantiated this protocol once as a frozen Direction-C search batch on 2026-05-26. The batch used seven fixed GitHub repository queries and five arXiv API metadata queries. It recorded direct hits, empty result sets, duplicate known surfaces, metadata-only paper rows, and one noisy title-query result before drawing any aggregate claim. The outcome is intentionally modest: metadata discovery is easy, but no new row-bound public target/split/score or response packet appeared. This strengthens the paper’s claim-support argument without pretending that a small metadata batch estimates field-wide artifact prevalence.

Figure 2 makes the corpus boundary visible. The v1 evidence-note corpus contains several scoreable or replayable

Experiment or public artifact → target/split check → row-level score or response packet → metric replay → evidence state → consumer boundary

Admitted rows can flow to audit reports. Candidate, support-only, watch, and negative rows remain in Research unless a later review changes the contract.

Fig. 1. DiffAudit evidence-contract pipeline. The pipeline is intentionally claim-oriented: the same metric can become admitted evidence, a candidate case study, support-only context, or a bounded negative depending on provenance and consumer semantics.

TABLE IV

CANDIDATE AND SUPPORT EVIDENCE GATE MATRIX. ‘PASS’ MEANS SUFFICIENT FOR THE STATED INTERNAL EVIDENCE ROLE, NOT AUTOMATIC PRODUCT ADMISSION. SURFACE DELTA IS EVALUATED FOR PORTABILITY OR SECOND-ASSET CLAIMS; THE ADMITTED BUNDLE IS THE PRIMARY AUDIT BUNDLE RATHER THAN A SECOND-ASSET CLAIM.

Route	Target	Split	Evidence	Metric	Boundary	Delta	Paper use
Admitted bundle	Pass	Pass	Pass	Pass	Pass	N/A	Main audit rows
H2 output-cloud	Pass	Pass	Pass	Pass	Partial	Partial	Candidate mechanism
H2 img2img portability	Pass	Pass	Pass	Pass	Fail	Pass	Portability limit
ReDiffuse STL-10 scout	Pass	Pass	Pass	Pass	Fail	Pass	Bounded negative
CommonCanvas response packet	Partial	Pass	Pass	Pass	Fail	Pass	Bounded negative
Tracing Roots feature packet	Partial	Pass	Pass	Pass	Partial	Pass	Support-only
SD ReDiffuse collaborator	Partial	Partial	Pass	Pass	Fail	Pass	Source-confounded support

TABLE V
METADATA-ONLY CORPUS EXPANSION PROTOCOL.

Field	Rule
Search cutoff	Use an absolute date before outcome coding.
Sources	arXiv, OpenReview, venue pages, GitHub, Zenodo, Hugging Face, Papers with Code.
Keywords	Diffusion MIA, generative MIA, diffusion memorization, data identification, synthetic-data MIA.
Recorded metadata	Paper URL, artifact URL, modality, target type, artifact type, public surface, gate labels.
Execution policy	No large downloads or GPU runs during metadata coding.

TABLE VI
FROZEN FIXED-SEARCH BATCH USED FOR DIRECTION-C CLAIM SUPPORT.

Recorded outcome	Rows
Direct GitHub/arXiv surfaces already covered by prior gates	3
New code-public surface with no audit-ready packet	1
Relevant arXiv metadata-only paper rows	8
Empty fixed-query repository result sets	4
Query-noise example retained for auditability	1
New admitted audit rows	0

rows, but most of those rows still stop at candidate, support-only, or negative roles once the consumer-boundary gate is applied. The fixed-search batch is stricter evidence against accidental promotion: metadata search found relevant papers and code surfaces, but no new row-bound public artifact that could become an admitted audit row. These counts are used only to support the paper’s claim-control argument. A selected-corpus consistency pass found no CSV label changes, but a standalone artifact-reproducibility paper would still need a

larger frozen corpus or a second independent label review before making aggregate claims.

B. Artifact Strata

We separate artifact strata before drawing conclusions. Admitted controls show what consumer-ready evidence looks like. Research candidates show where a strong signal exists but portability is open. Positive non-admitted packets show that scoreability and auditability differ. Bounded negatives show which routes are not worth expanding under the current hypothesis. Source-confounded packets show why row semantics cannot be inferred from a headline AUC. This stratification prevents two common errors: treating all public artifacts as failed audits, and treating all positive scores as portable audit evidence.

VII. ADMITTED EVIDENCE BUNDLE

Rows are admitted through a fixed path. First, a score array or response packet must bind every sample to a member/nonmember label under a stated target. Second, the metrics in Table VII are replayed from those rows or from a frozen response scorer. Third, low-FPR readouts are attached to their finite nonmember denominator. Finally, the row receives consumer-boundary language that states who may use it and what caveat must accompany it. This is why admitted rows can coexist with stronger non-admitted candidates: admission is about reusable audit semantics, not only metric magnitude.

Table VII summarizes the admitted rows. Each row is consumed with its cost and limitation language; none is treated as a universal benchmark.

VIII. OUTPUT-CLOUD GEOMETRY CASE STUDY

The H2 response-strength cache contains repeated diffusion responses at timesteps 40, 80, 120, and 160. Instead of comparing a query image directly to outputs, the output-cloud scorer uses only geometry among outputs: within-step repeat

Artifact corpus gate labels (counts, not prevalence estimates) Partial Fail

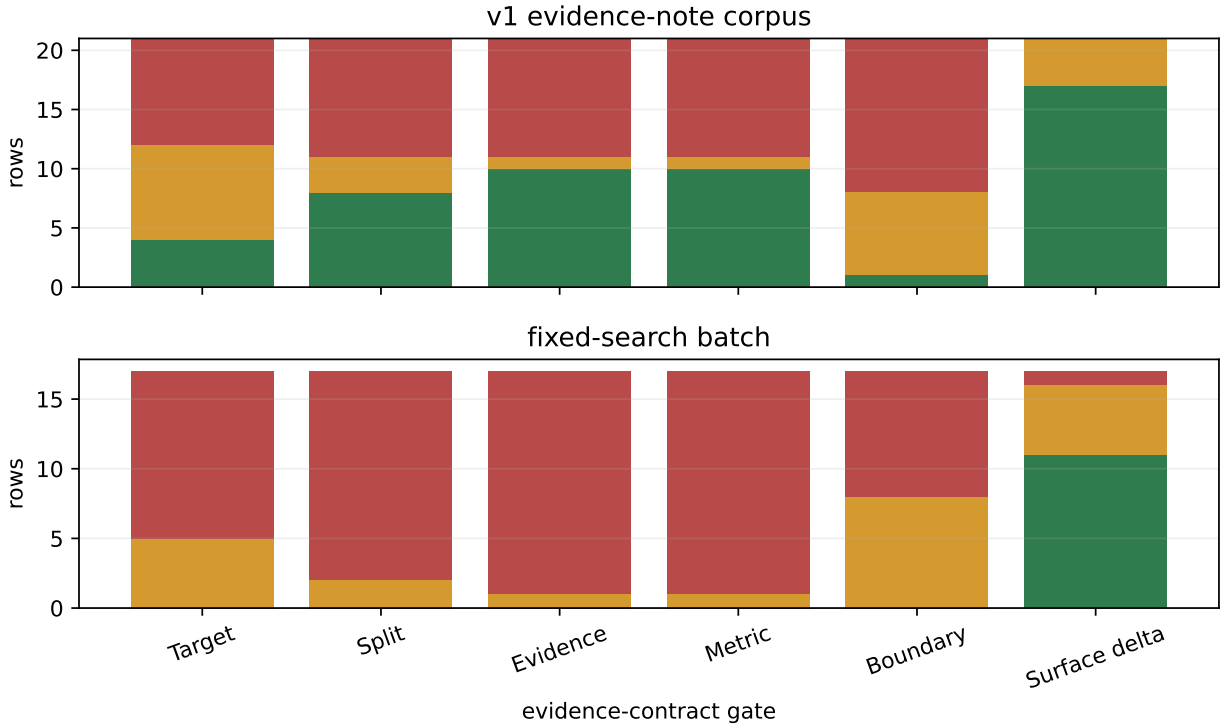


Fig. 2. Gate-label counts for the selected v1 evidence-note corpus and the frozen 2026-05-26 fixed-search batch. Counts describe coded rows in these two corpora only; they are not field-wide prevalence estimates. The fixed-search batch produced no row passing the target, split, evidence, metric, or consumer boundary gate, so it supports claim control rather than a standalone reproducibility-prevalence claim.

TABLE VII
ADMITTED DIFFAUDIT EVIDENCE ROWS.

Track	Method	AUC	ASR	TPR@1%FPR	TPR@0.1%FPR	Role
Black-box	recon DDIM public-100 step30	0.837	0.740	0.220	0.110	controlled risk proof
Gray-box	PIA GPU512 baseline	0.841	0.786	0.059	0.012	baseline
Gray-box	PIA + stochastic dropout	0.828	0.768	0.053	0.010	provisional comparator
White-box	GSA 1k-3shadow	0.998	0.990	0.987	0.432	upper-bound attack
White-box	GSA against DPDM W-1	0.489	0.499	0.009	0.000	defense comparator

RMSE, cross-timestep centroid RMSE, and small response-cloud PCA/Gram features. This design asks whether the response cloud itself carries membership information.

On the existing 512/512 cache, output-cloud logistic scoring reaches AUC 0.9615, ASR 0.9004, TPR@1%FPR 0.3340, and TPR@0.1%FPR 0.1172. It improves on raw H2 logistic by 0.0558 AUC and on lowpass H2 logistic by 0.0659 AUC. A label shuffle returns random-level AUC 0.5076. Because the historical cache used class-ordered seed offsets, DiffAudit released only bounded 256/256 shared-position controls: with identical per-position seed offsets for members and nonmembers, the scorer still reaches AUC 0.9678 at seed 176 and 0.9562 at seed 177. A fold-disjoint same-family cross-cache transfer review gives AUC 0.9490 from seed 176 to 177 and

AUC 0.9705 from seed 177 to 176.

The portability boundary is equally important. On the SD/CelebA img2img admission cache, output-cloud geometry reaches only AUC 0.7888 with zero strict tail recovery and underperforms the existing best simple-distance comparator by 0.088 AUC. A smaller stability cache is positive but still below simple distance. Thus the paper reports output-cloud geometry as a controlled mechanism candidate, not as an admitted black-box row.

IX. NEGATIVE AND SUPPORT EVIDENCE

DiffAudit treats failed routes as measurement evidence when they answer a specific decision question. ReDiffuse STL-10 provides exact public split manifests and a locally

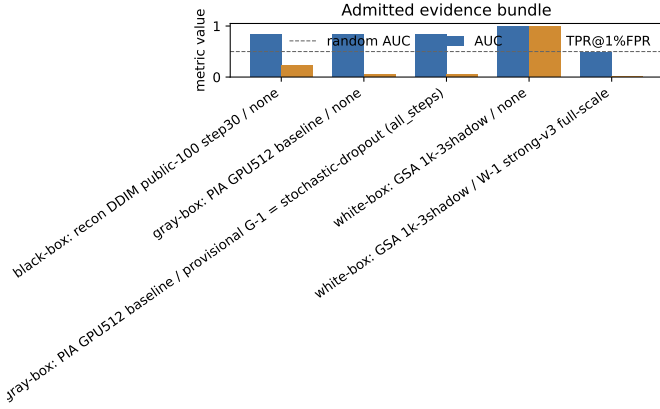


Fig. 3. AUC and TPR@1%FPR for admitted rows. The figure compares evidence under different access and defense contracts; it is not a single benchmark leaderboard.

TABLE VIII
WHY H2 OUTPUT-CLOUD GEOMETRY IS NOT ADMITTED.

Check	Measurement decision
H2 main cache	Strong positive candidate signal.
Label shuffle	Random-level sanity check supports nontrivial signal.
Shared-position controls	Same-family seed-offset explanation is insufficient.
Cross-cache transfer	Same-family robustness, not cross-model portability.
SD/CelebA img2img	Weak or not distinct from simple distance.
Admission decision	Research candidate only; no product/runtime row.

scoreable bounded target, but fixed-timestep denoising-loss is random-level (AUC 0.4996) and a SimA-style score-norm observable remains weak (AUC 0.5053), even though ReDiffuse is a relevant black-box diffusion MIA reference point [12]. CommonCanvas/CopyMark has a real 50/50 response contract, yet conditional denoising-loss reaches only AUC 0.5148. Our replay of the Tracing the Roots feature tensors is positive (AUC 0.8158) [16], but they are not raw image-level evidence and lack target checkpoint/sample regeneration artifacts for the current consumer boundary. The collaborator Stable Diffusion ReDiffuse packet is replayable (AUC 0.7103), but source labels alone separate members and nonmembers perfectly, so it is a cross-source stress test rather than same-distribution membership evidence.

X. DISCUSSION AND LIMITATIONS

The strongest criticism of DiffAudit is that evidence contracts may look like engineering governance rather than research. We argue the opposite: in diffusion membership inference, evidence contracts are part of the measurement method. Without them, a high AUC can come from a source-label shortcut, a prompt-conditioned auxiliary feature, a non-replayable score array, or a finite-tail artifact. The H2 case study shows that DiffAudit can still discover new technical

signals; the contract simply controls how far those signals are allowed to travel. This position is also consistent with broader caution that membership inference alone should not be overstated as definitive proof of training-data use [9].

This draft does not claim broad generalization across all diffusion models. It does not claim that H2 output-cloud geometry is product-ready. It does not claim that bounded weak scouts disprove the original methods they resemble. The next scientific step for a full CCF-B+ submission is to add either a second checkpoint-bound response asset for the output-cloud mechanism or a larger, pre-registered public-artifact audit corpus for the evidence-contract methodology.

A. Why Contracts Are Measurement

The central methodological claim is that membership leakage is not measured by a metric alone. It is measured by a metric together with the identity and provenance conditions that make the metric meaningful. This is especially important for diffusion models, where the same paper may involve a base model, fine-tuned checkpoint, prompt set, stochastic generation policy, intermediate trajectory tensor, and several possible member definitions. A score detached from these conditions cannot answer the audit question even if it is statistically strong.

DiffAudit therefore treats the evidence contract as an experimental control. It is analogous to fixing a split, declaring a threat model, or publishing a random seed: the contract restricts what can be inferred from the observation. The H2 result illustrates the payoff. The output-cloud signal is strong enough to motivate a mechanism paper, but the contract prevents it from becoming a runtime claim until a second response asset or a successful portability check exists. Conversely, the admitted GSA row is extremely strong, but the contract keeps it an upper-bound white-box comparator rather than a black-box audit result.

B. Artifact Release Checklist

The following release recommendations are derived from the observed gate failures in the v1 corpus, the fixed-search batch in Table VI, and the decision-value cases in Table IX. For diffusion MIA results to become reusable audit evidence, future artifacts should publish more than code and headline tables. A reusable packet should include: target model and checkpoint identity; immutable member and nonmember manifests; row-level scores or response manifests; metric code or metric JSON; nonmember denominators for low-FPR tails; artifact hashes or repository commit IDs; and a statement of the intended consumer boundary. If a result depends on intermediate trajectories, feature tensors, prompt sources, or regenerated samples, those surfaces should be named explicitly. This checklist is not meant to penalize prior work; it describes the minimum information needed for a third party to decide whether a score supports an audit claim, a mechanism claim, or only a paper-local comparison.

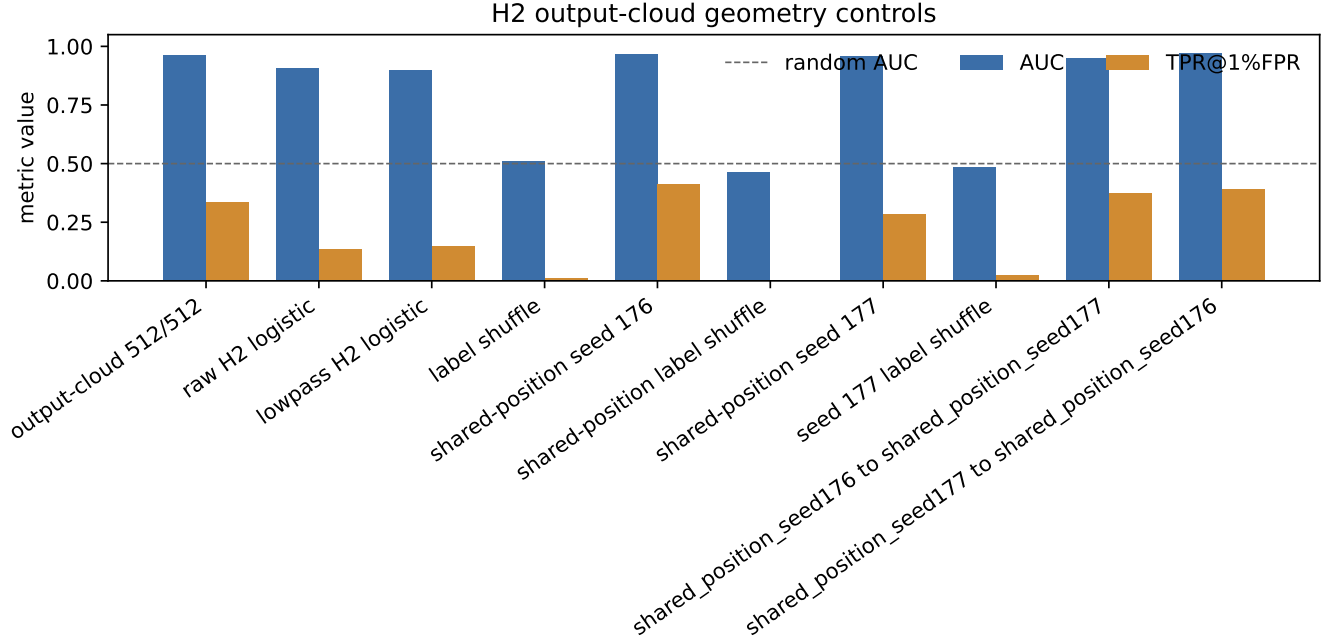


Fig. 4. H2 output-cloud geometry controls. The signal beats raw/lowpass baselines and survives label-shuffle, seed-offset control, seed stability, and same-family cross-cache transfer. It remains a research candidate because portability to the SD/CelebA img2img contract is weak or not distinct from simple distance.

TABLE IX
DECISION VALUE OF NON-ADMITTED EVIDENCE.

Route	Decision question	Overclaim blocked
ReDiffuse STL-10	Does a clean split plus bounded denoising-loss/score-norm scout release a larger run?	It does not justify a same-family expansion or a failed-reproduction claim.
CommonCanvas response packet	Does a second response contract support current scorers?	A real response packet alone is not a portability result.
MIDST Blending++	Does tabular diffusion evidence reopen an image/latent audit row?	Cross-domain scoreability is not an admitted image-diffusion claim.
Tracing Roots feature tensors	Does a positive feature packet support raw image audit consumption?	Feature-level replay is not a raw checkpoint/sample audit.
SD ReDiffuse collaborator packet	Does replayability imply same-distribution membership evidence?	Source confounding blocks a same-distribution claim.
Fixed-search batch	Does metadata discovery produce a new admitted row?	Small metadata search is not field-wide artifact prevalence.

C. Threats to Validity

Internal validity. Some results are produced by local replay scripts and repository artifacts rather than by re-running complete upstream training pipelines. DiffAudit mitigates this by recording row-level provenance and by refusing to call bounded scouts full reproductions.

External validity. The admitted bundle is permission- and asset-bound. It should not be read as evidence that every diffusion model leaks at the same level, nor that one method dominates across modalities.

Metric validity. Low-FPR tails can be misleading on small packets. We therefore report them as finite empirical readouts and keep packet size and nonmember denominators attached to the claim.

Publication bias. Negative routes were selected because

they affected DiffAudit route decisions. The 2026-05-26 fixed-search batch reduces project-history bias, and the selected-corpus consistency pass catches internal label inconsistencies, but both remain single-review, metadata-light process traces rather than prevalence estimates. A standalone artifact-claim paper would need a larger frozen corpus or a second independent label review before drawing broader conclusions about public artifact availability.

XI. CONCLUSION

DiffAudit turns diffusion membership inference from isolated score reporting into evidence-contracted auditing. The current results show verified leakage under admitted black-box, gray-box, and white-box contracts; reveal a promising output-cloud response geometry signal; and document why

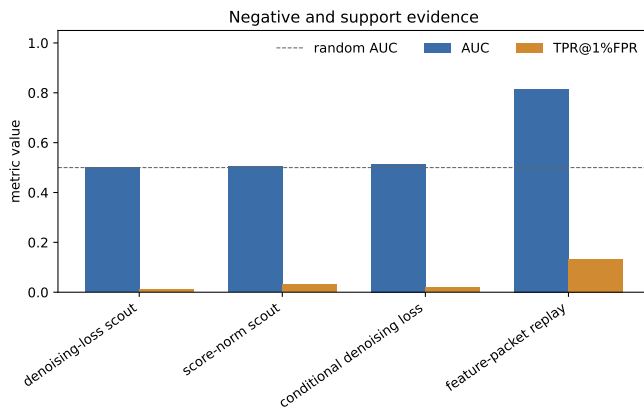


Fig. 5. Negative and support evidence. These results are not failures of the audit framework; they are examples of why scoreability, portability, and consumer semantics must be separated.

many public artifacts remain scientifically useful but non-admissible. This combination is the paper’s central point: publishable diffusion privacy evidence requires both strong signals and strict claim boundaries.

REFERENCES

- [1] J. Duan, F. Kong, S. Wang, X. Shi, and K. Xu, “Are diffusion models vulnerable to membership inference attacks?” in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023. [Online]. Available: <https://proceedings.mlr.press/v202/duan23b.html>
- [2] F. Kong, J. Duan, R. Ma, H. T. Shen, X. Shi, X. Zhu, and K. Xu, “An efficient membership inference attack for the diffusion model by proximal initialization,” in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=rpH9FcCEV6>
- [3] S. Zhai, H. Chen, Y. Dong, J. Li, Q. Shen, Y. Gao, H. Su, and Y. Liu, “Membership inference on text-to-image diffusion models via conditional likelihood discrepancy,” in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: <https://openreview.net/forum?id=DztaBt4wP5>
- [4] Y. Pang and T. Wang, “Black-box membership inference attacks against fine-tuned diffusion models,” in *Network and Distributed System Security Symposium*, 2025. [Online]. Available: <https://www.ndss-symposium.org/wp-content/uploads/2025-324-paper.pdf>
- [5] Y. Pang, T. Wang, X. Kang, M. Huai, and Y. Zhang, “White-box membership inference attacks against diffusion models,” in *Proceedings on Privacy Enhancing Technologies*, vol. 2025, no. 2, 2025, pp. 398–415.
- [6] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, “Membership inference attacks against machine learning models,” in *2017 IEEE Symposium on Security and Privacy*, 2017, pp. 3–18.
- [7] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, “Privacy risk in machine learning: Analyzing the connection to overfitting,” in *2018 IEEE 31st Computer Security Foundations Symposium*, 2018, pp. 268–282.
- [8] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, “Membership inference attacks from first principles,” in *2022 IEEE Symposium on Security and Privacy*, 2022, pp. 1897–1914.
- [9] J. Zhang, D. Das, G. Kamath, and F. Tramèr, “Position: Membership inference attacks cannot prove that a model was trained on your data,” in *2025 IEEE Conference on Secure and Trustworthy Machine Learning*, 2025, pp. 333–345.
- [10] N. Carlini, J. Hayes, M. Nasr, M. Jagielski, V. Schwag, F. Tramèr, B. Balle, D. Ippolito, and E. Wallace, “Extracting training data from diffusion models,” in *Proceedings of the 32nd USENIX Security Symposium*, 2023, pp. 5253–5270.
- [11] G. Somepalli, V. Singla, M. Goldblum, J. Geiping, and T. Goldstein, “Diffusion art or digital forgery? investigating data replication in diffusion models,” in *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 6048–6058.
- [12] J. Li, J. Dong, T. He, and J. Zhang, “Towards black-box membership inference attack for diffusion models,” in *Proceedings of the 42nd International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 267. PMLR, 2025, pp. 34 271–34 287. [Online]. Available: <https://proceedings.mlr.press/v267/li25k.html>
- [13] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d’Alché Buc, E. Fox, and H. Larochelle, “Improving reproducibility in machine learning research (a report from the neurips 2019 reproducibility program),” *Journal of Machine Learning Research*, vol. 22, no. 164, pp. 1–20, 2021. [Online]. Available: <https://www.jmlr.org/papers/v22/20-303.html>
- [14] T. Gebru, J. Morgenstern, B. Vecchione, J. W. Vaughan, H. Wallach, H. Daumé III, and K. Crawford, “Datasheets for datasets,” *Communications of the ACM*, vol. 64, no. 12, pp. 86–92, 2021.
- [15] M. Mitchell, S. Wu, A. Zaldivar, P. Barnes, L. Vasserman, B. Hutchinson, E. Spitzer, I. D. Raji, and T. Gebru, “Model cards for model reporting,” in *Proceedings of the Conference on Fairness, Accountability, and Transparency*, 2019, pp. 220–229.
- [16] A. Floros, S.-M. Moosavi-Dezfooli, and P. L. Dragotti, “Tracing the roots: Leveraging temporal dynamics in diffusion trajectories for origin attribution,” 2025. [Online]. Available: <https://arxiv.org/abs/2411.07449>